Android Introduction

LEIM – DAM

What is Android?

- A Open source platform for touchscreen mobile devices
- A Linux-kernel based operating system
- Created by Andy Rubin (Android Inc.) in 2003
- Developed by the Open Handset Alliance, led by Google

- "Android is a mobile operating system based on a modified version of the Linux kernel and other open source software, designed primarily for touchscreen mobile devices such as smartphones and tablets. Android is developed by a consortium of developers known as the Open Handset Alliance, with the main contributor and commercial marketer being Google." Wikipedia

Android history

Android logo 2007–2014

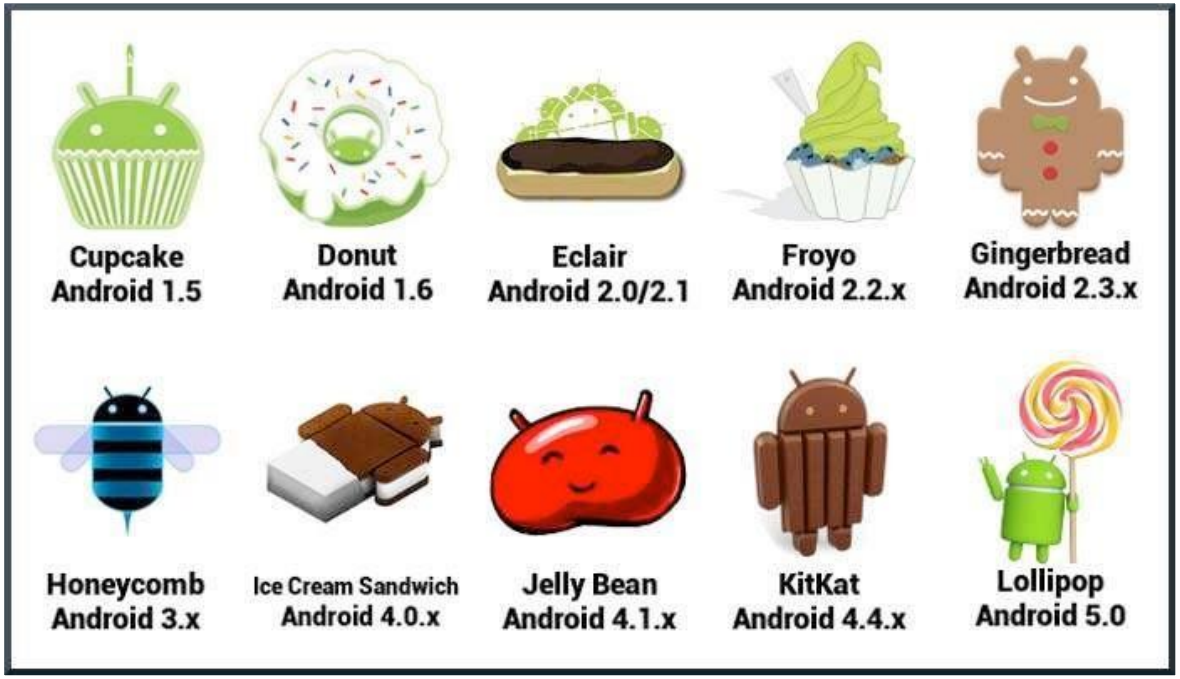
Android logo 2014–2019

- 2003, Android Inc. founded by Andy Rubin, Rich Miner, Nick Sears and Chris White
- 2005, Google acquires Android Inc.
- 2007, Open Handset Alliance (OHA) is formed
- 2007, Beta version of the Android SDK
- 2008, **Android 1.1** version in a mobile device HTC DREAM (T-Mobile G1)
- 2010, Android **Eclair** (2.0) Android is the mobile platform with the largest number of active devices
- 2011, Android **Honeycomb** (3.x) tablets only
- 2011, Android **Ice Cream Sandwich** (4.0)
- 2012, Android Jelly Bean (4.1)
- 2013, Android KitKat (4.4)
- 2014, Android **Lollipop** (version 5) - Android Runtime (ART) and support for 64-bit architectures
- 2015, Android **Marshmallow** (version 6) – USB-C, Fingerprint authentication, Android Pay
- 2016, Android **Nougat** (version 7) – Improved multitasking, multi-window mode, support for virtual reality
- 2017, Android **Oreo** (version 8) – Picture-in-Picture, APIs for Artificial Intelligence and shared memory
- 2018, Android **Pie** (version 9) – updated user interface, richer notifications, biometric authentication

Android history

- 2019, Android **10**, Gesture navigation; system-wide dark theme; major privacy controls
- 2020, Android **11**, Conversations & chat bubbles; one-time permissions
- 2021, Android **12**, Material You dynamic theming; major UI redesign; privacy dashboard
- 2022, Android **12L**, Large-screen optimization; taskbar & improved multitasking
- 2022, Android **13,** Per-app language; notification permission; themed icons
- 2023, Android **14**, Accessibility improvements; lock-screen customization; predictive back gesture
- 2024, Android **15**, Private Space; enhanced privacy sandbox; foldable continuity
- 2025/2026, Android **16**, Expected: deeper on-device AI; XR/large-screen/desktop experiences

Logos and Names of Android Versions



Marshmallow
Android 6

Nougat
Android 7

Oreo
Android 8

Pie
Android 9

android 10

Q
Android 10

R
Android 11

Android version

Platform Version	API Level	Version Code
Android 11.0	R	R
Android 10.0	29	Q
Android 9.0	28	Pie
Android 8.0 / 8.1	26 / 27	Oreo
Android 7.0 / 7.1.1	24 / 25	Nougat
Android 6.0	23	Marshmallow
Android 5.1	22	Lollipop_MR1
Android 5.0	21	Lollipop
Android 4.4W	20	KitKat_Watch
Android 4.4	19	KitKat
Android 4.3	18	Jelly_Bean_MR2
Android 4.2, 4.2.2	17	Jelly_Bean_MR1
Android 4.1, 4.1.1	16	Jelly_Bean
Android 4.0.3, 4.0.4	15	Ice_Cream_Sandwich_MR1
Android 4.0, 4.0.1, 4.0.2	14	Ice_Cream_Sandwich
…	…	…

Versions increment name first letter

Market use

- Target platform – best practice
  - An app should target at least 90% of devices

ANDROID PLATFORM VERSION	API LEVEL	CUMULATIVE DISTRIBUTION
4.0 Ice Cream Sandwich	15	
4.1 Jelly Bean	16	99,6%
4.2 Jelly Bean	17	98,1%
4.3 Jelly Bean	18	95,9%
4.4 KitKat	19	95,3%
5.0 Lollipop	21	85,0%
5.1 Lollipop	22	80,2%
6.0 Marshmallow	23	62,6%
7.0 Nougat	24	37,1%
7.1 Nougat	25	14,2%
8.0 Oreo	26	6,0%

Extract from Android Studio on March, 2020

Android architecture

User apps

System apps providing base services for user and other apps.

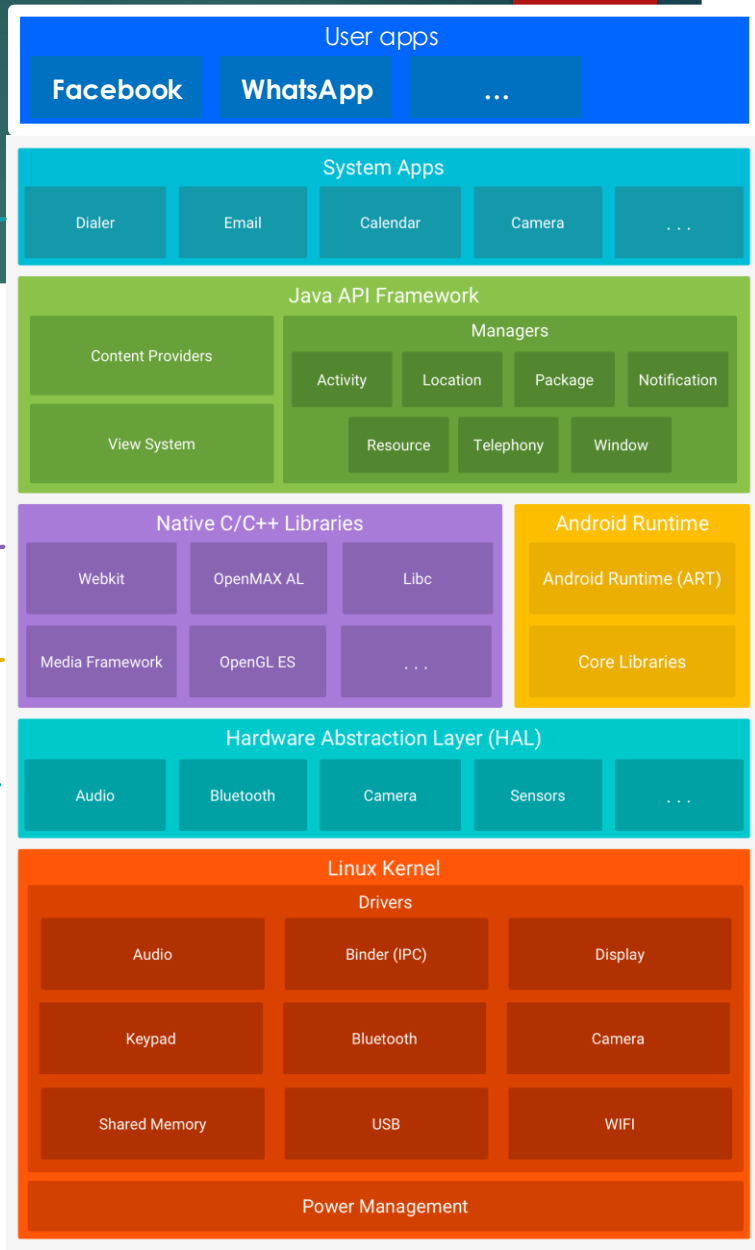
Java API providing the environment and resources to Android App development.

Environment to access and develop to/of **native code (C/C++)**.

Runtime to execute Android apps.

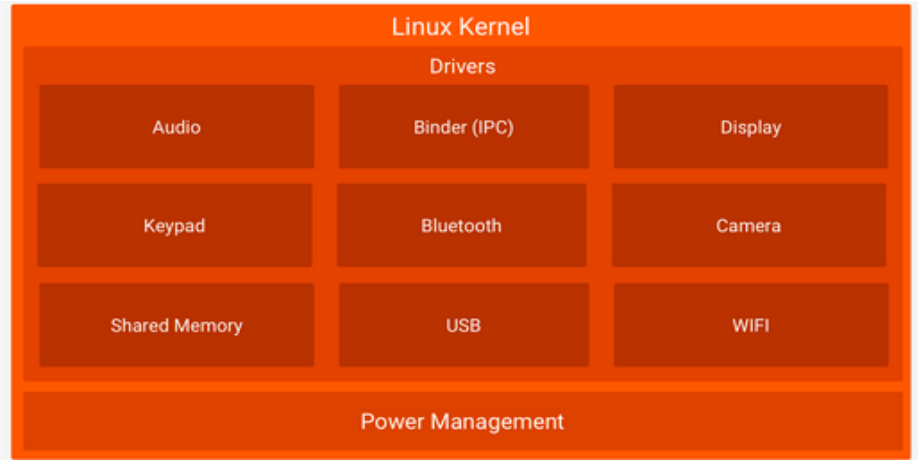
Abstraction layer providing a uniform way to access to device hardware capabilities (Bluetooth, camera, …)

Linux Kernel providing base OS services



Architecture: Linux Kernel

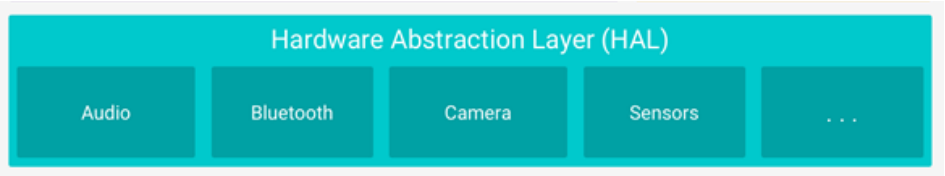
- Linux Kernel provides the base OS services
  - (processes, threads, memory, file system, …).
- Changed with:
  - wakeLocks,
  - aggressive memory management,
  - Inter-Process Communication (Binder) that minimizes memory use.
- Allows manufacturers to develop hardware drivers for Linux.



https://developer.android.com/guide/platform

Architecture: abstraction layer

- **Abstraction layer** provides a uniform way (API) to access to device hardware capabilities
  - Bluetooth
  - Camera
  - Sensors

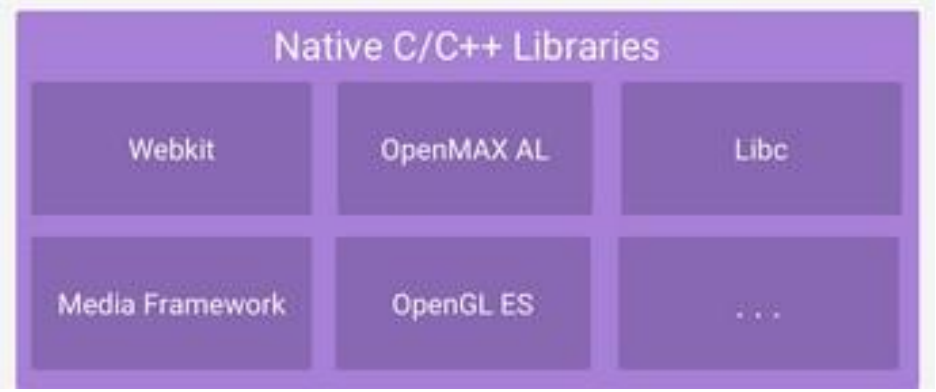


- activity_recognition.h
- audio.h
- audio_alsaops.h
- audio_effect.h
- audio_policy.h
- bluetooth.h
- boot_control.h
- camera.h
- camera2.h
- camera3.h
- camera_common.h
- consumerir.h
- context_hub.h
- fb.h
- fingerprint.h

https://android.googlesource.com/platform/hardware/libhardware/+/master/include/hardware

Architecture: Native Libraries

- Exposes **native libraries in C/C++**
  - In Java framework APIs
  - To access: 2D and 3D graphic libraries
- Native Development Kit (NDK) **to code in C/C++**



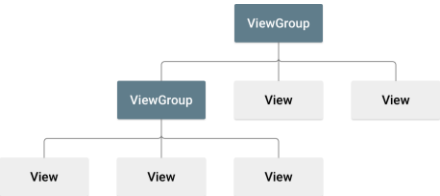
Architecture: Android Runtime

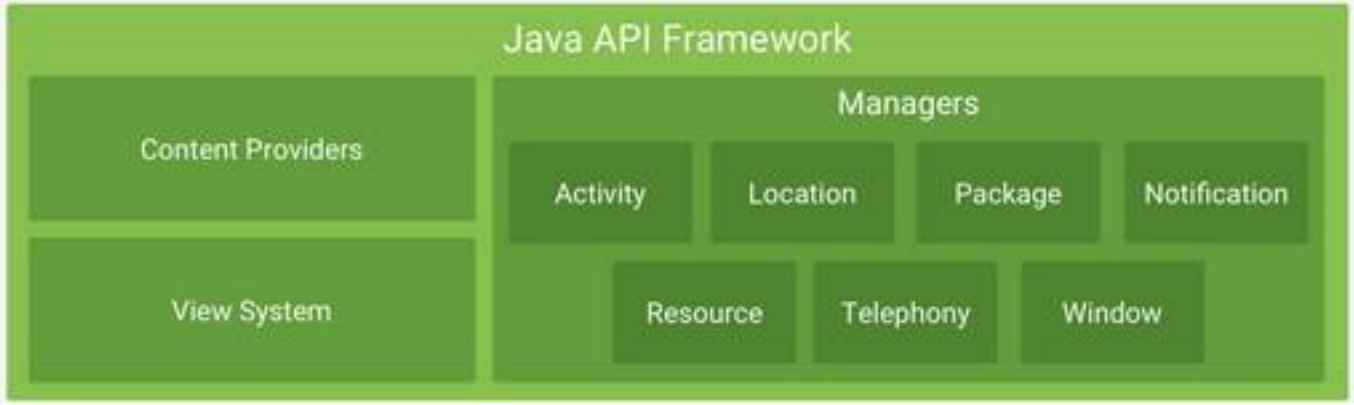
- **Android Runtime (ART)** (Dalvik Virtual Machine for Android < 5.0)
  - Runtime to execute .dex files
    - similar to a JVM, but optimized for Android
    - .dex files are more compact and efficient than class files
  - Suitable for limited memory and battery systems
  - An app runs on a process in a dedicated instance of the Android Runtime (ART).
  - ART platform may run multiple ART instances.

- Core Libraries
  - Java core Libraries

Architecture: Java API Framework

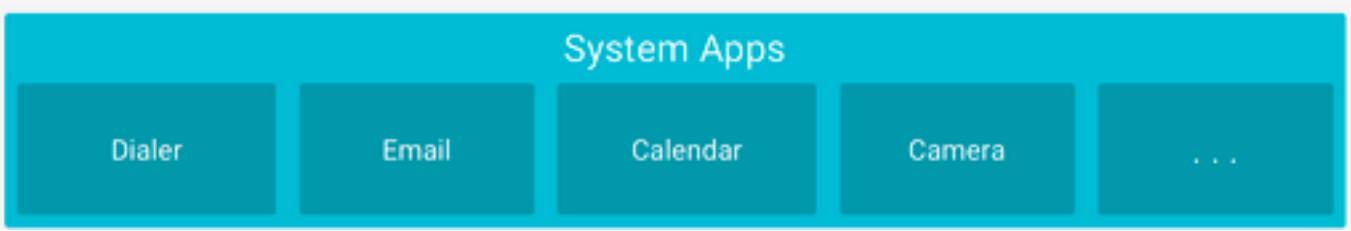
- Java API to all Android OS
- Provides the environment for Android App development
  - Activity manager: manages activities life cycle
  - Resource manager: access to embedded resources (Strings, colors, images, interfaces screens, …)
  - Content providers: to publish data and access data from other apps
  - View System: a system and a set of resources to develop User Interfaces





Architecture: System Applications

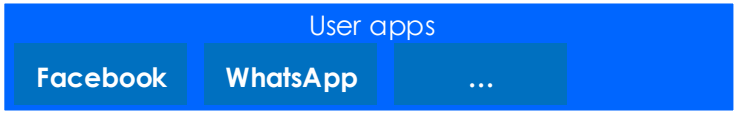
- System apps
  - Have access to all platform APIs
  - Built-in (Android or manufacturer) applications
  - Provide base services (services and content providers) for user and other apps
    - Ex: Camera, Messages, Bluetooth, Calendar, Phone, Music, …
  - Placed on: /system/app (/app)
  - To see these apps: adb shell pm list packages -s



https://android.jlelse.eu/system-app-in-android-f003d418b4cc

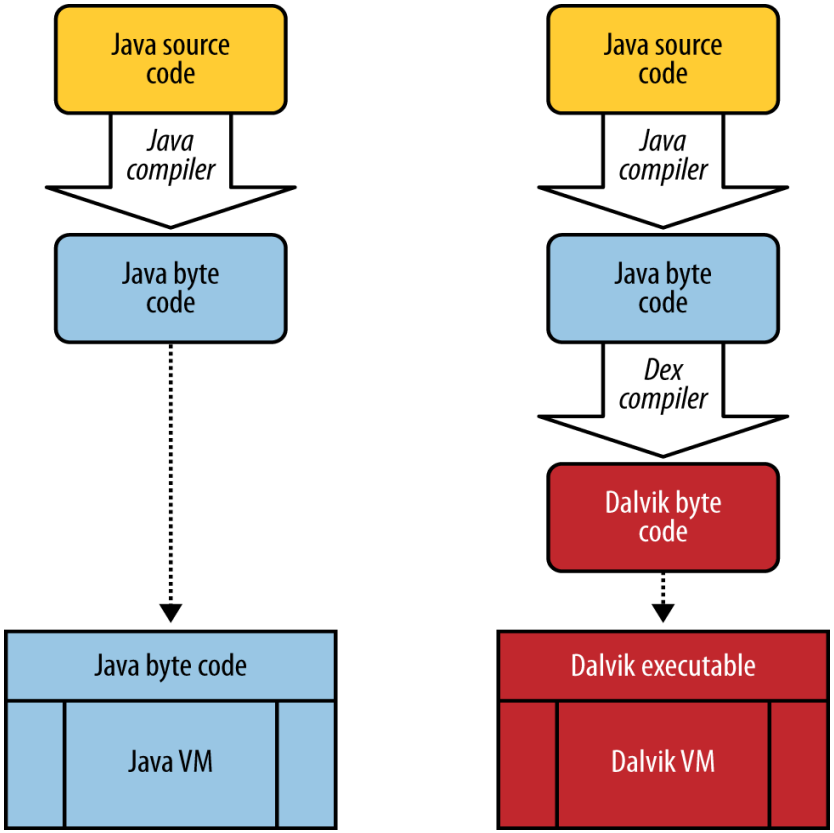
Architecture: User Applications

- User apps
  - Apps installed by Google PlayStore and by user direct actions
  - Placed on: /data/app
  - To see these apps: adb shell pm list packages -3

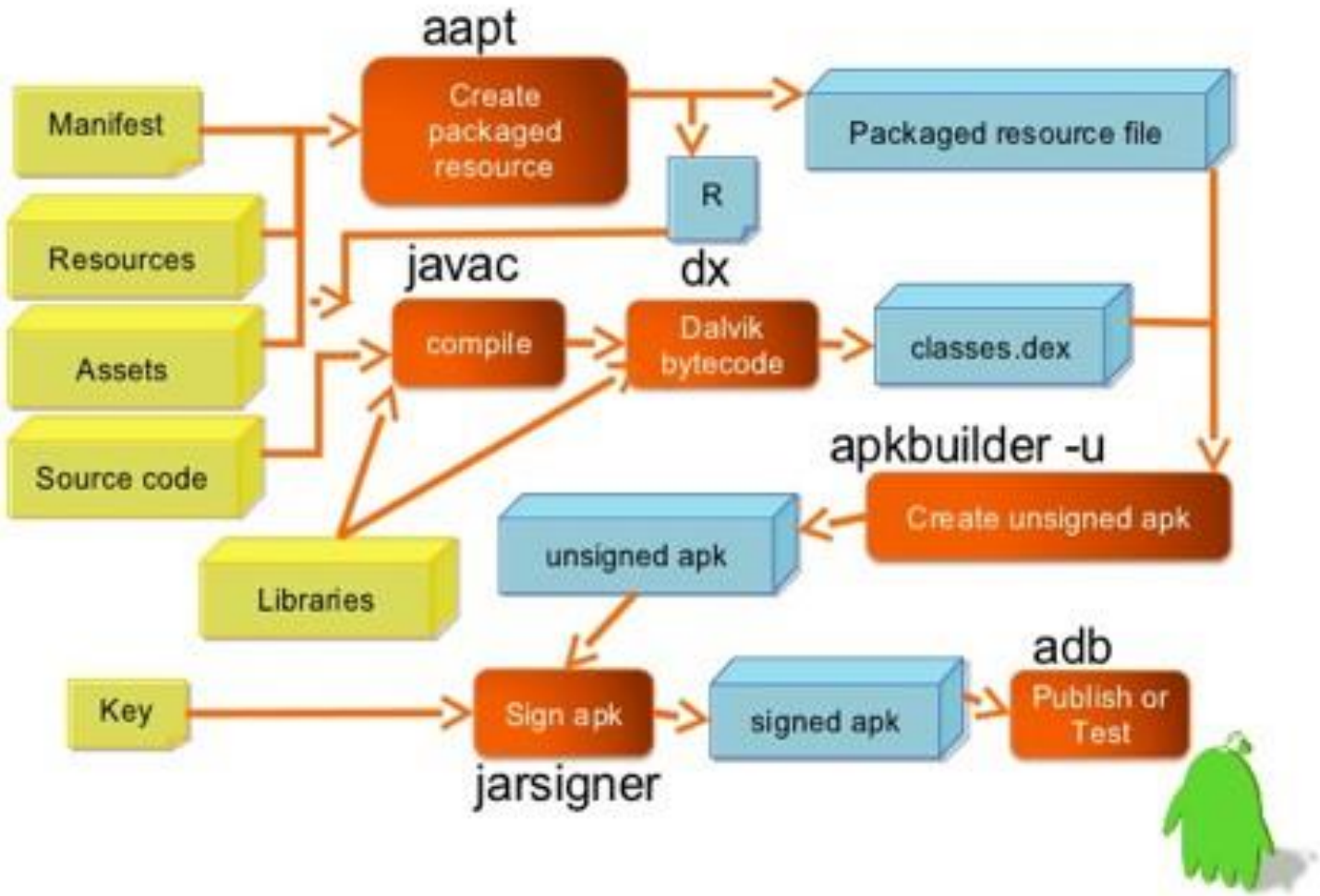


Runtime: Java and Android

- Java
  - Compile .java to .class files
    - Java bytecode
  - Group .class in .jar files
  - JVM excutes .class files
- Android
  - Compile **.java** to .class files
  - Compile .class to **.dex** files
    - Dex/Dalvik bytecode
    - Dex – Dalvik executable
  - Grouped in an **.apk** file
  - Dalvik/ART executes dex code

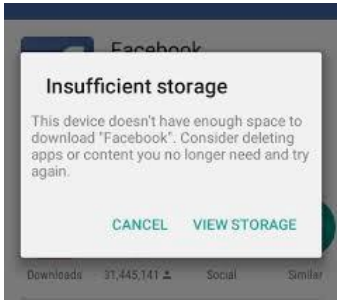


Android application development and deployment flow



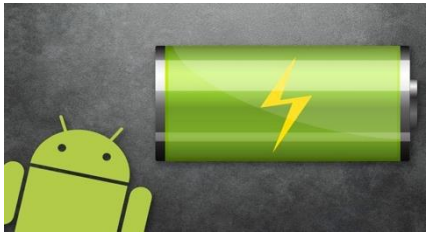
Dalvik bytecode

- "Dalvik" bytecode – Dalvik Executalbe format - **.dex** files
  - Dalvik and ART platforms run .dex files
- Programs are written in Java compiled for Java bytecode and converted to "Dalvik" bytecode
  - Or written in Kotlin
- Multiple classes are included in a single .dex file.
- An app is packed in an **.apk** file (Android Package Kit file)
- Strings in all used classes are included only once in .dex file to conserve space
  - No duplication



Android Runtime (ART)

- **ART: Android Runtime** environment
  - replaces Dalvik virtual machine from Android 5.0
  - ART uses Dalvik byte code
  - Art uses ahead-of-time (AOT) compilation
    - It compiles the app bytecode to native code once at app installation
      - Dalvik uses Just-in-Time compilation, compiling the app bytecode every time that is used
    - Better application performance, less processor and battery use
      - But with greater installation time and ART files need some more space
    - ART improves (over Dalvik) memory allocation, Garbage Collection and allows better debugging and profiling

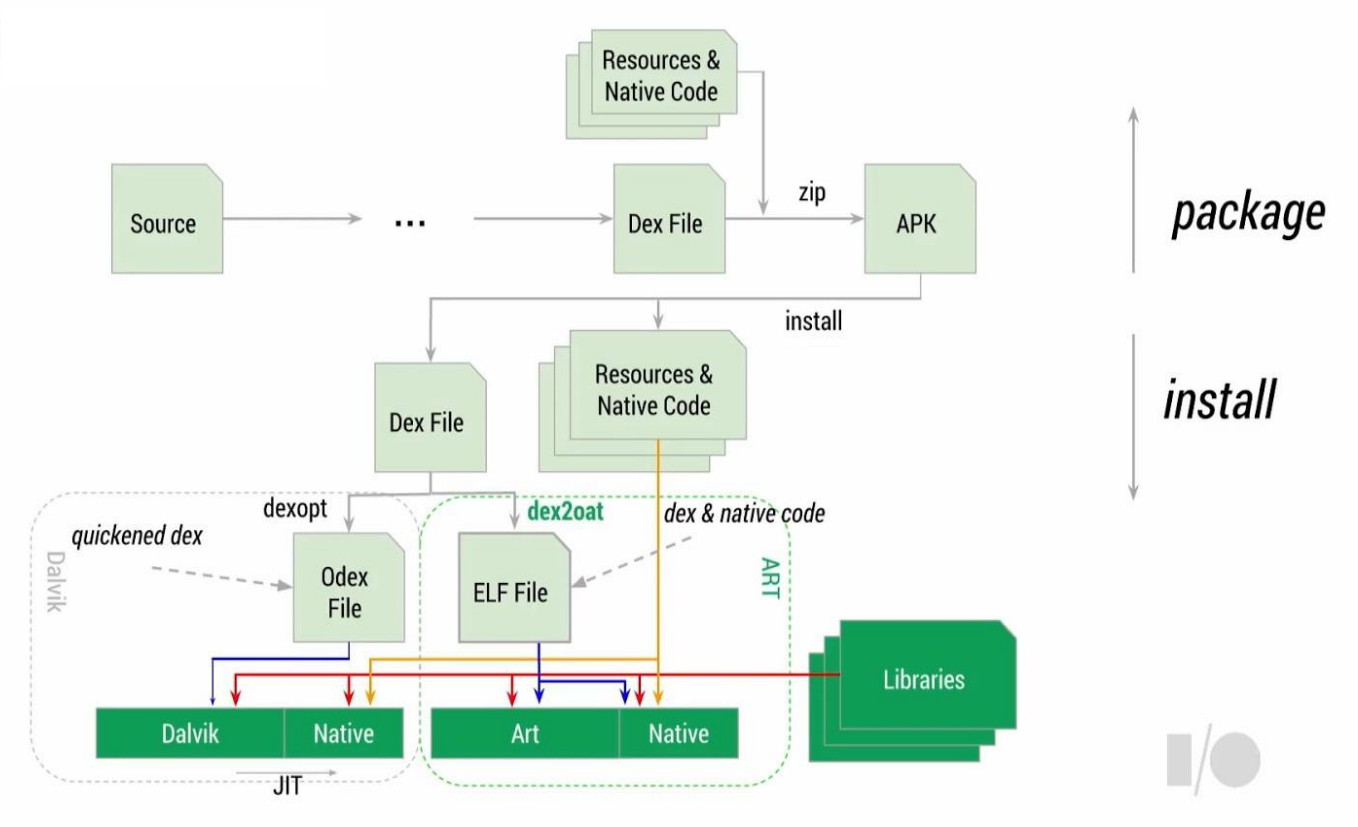


https://source.android.com/devices/tech/dalvik/index.html

Android Application Package (APK)

- An .APK file is a zip file containing
  - **Android Manisfest file**
    - defines app componentes and settings, like permissions, Android version needed, ...
  - **Dalvik executable** (.dex)
    - Java/Kotlin code compiled for .dex executable
  - **Resources**
    - everything that is not code
  - **Native libraries**
    - C / C ++ libraries
  - **Digital signature**
    - META-INF directory

ART and Dalvik: artifacts



- .odex – Optimized Dex file
- .elf – Executable and Linkable Format

Summary

- Android - an open source platform for mobile devices that is a Linux kernel-based operating system
- Layers of the Android environment - Linux Kernel, Hardware abstraction layer, Native C/C++ Libraries, Android Runtime, Java API Framework, System Apps
- Android Application Package (APK) – an APK containing all the files of one app
- Android Execution Runtime: ART and Dalvik, Ahead-of-Time (AOT) and Just-in-Time (JIT) compilation, ART versus Dalvik